

VPA IB DP CS — Bilingual Glossary (180 Terms)

VPA IB DP Computer Science — Bilingual Glossary 双语词汇表 Shuāngyǔ Cíhuì Biǎo

Total Terms: 180 | Language: English / 中文 / Pinyin

Usage: This glossary supports EAL students in all lessons, assessments, and the Internal Assessment. Students should maintain their own personal glossary with examples.

How to Use This Glossary

- Review 10 terms per week as part of homework
- Create flashcards (Anki or physical) for spaced repetition
- Write an example sentence using each term in context
- Use bilingual definitions in your IA if needed (with teacher guidance)

Complete Term List

#	English	中文	Pinyin	Definition
1	Algorithm	算法	suànfǎ	A step-by-step procedure for solving a problem
2	Array	数组	shùzǔ	A collection of elements of the same type, indexed by position
3	Binary	二进制	èrjìnzhì	A base-2 number system using digits 0 and 1
4	Binary search	二分搜索	èrfēn sōusuǒ	A search algorithm that repeatedly divides a sorted list in half
5	Binary search tree	二叉搜索树	èrchà sōusuǒ shù	A binary tree where left < parent < right
6	Binary tree	二叉树	èrchà shù	A tree where each node has at most two children
7	Bit	位	wèi	The smallest unit of data in computing (0 or 1)
8	Boolean	布尔	bù'ěr	A data type with two possible values: true or false

#	English	中文	Pinyin	Definition
9	Breadth-first search	广度优先搜索	guǎngdù yōuxiān sōusuǒ	Explores all neighbours at current depth before moving deeper
10	Bubble sort	冒泡排序	màopào páixù	A sorting algorithm that repeatedly swaps adjacent elements if in wrong order
11	Bus	总线	zǒngxiàn	A communication pathway for data between components
12	Byte	字节	zìjié	A group of 8 bits
13	Cache	高速缓存	gāosù huǎncún	Fast memory between CPU and RAM for frequently used data
14	Centralized control	集中式控制	jízhōngshì kòngzhì	A system where all decisions are made by a single controller
15	Central processing unit	中央处理器	zhōngyāng chǔlǐ qì	The main processing unit of a computer
16	Client-server	客户机-服务器	kèhùjī-fúwùqì	A network architecture where clients request services from central servers
17	Cloud computing	云计算	yún jìsuàn	Delivering computing services over the internet
18	Compression	压缩	yāsuō	Reducing file size by encoding data more efficiently
19	Concurrency	并发	bìngfā	Multiple computations executing simultaneously
20	Consistency	一致性	yīzhì xìng	All nodes in a distributed system seeing the same data
21	Control unit	控制单元	kòngzhì dānyuán	Directs operation of the processor
22	Database	数据库	shùjùkù	An organised collection of structured data
23	Data bus	数据总线	shùjù zǒngxiàn	Transfers data between processor, memory, and I/O
24	Deadlock	死锁	sǐsuǒ	A situation where processes are blocked waiting for each other
25	Decimal	十进制	shíjìnzhì	Base-10 number system

#	English	中文	Pinyin	Definition
26	Decision table	决策表	juécè biǎo	A table showing conditions and corresponding actions
27	Denary	十进制	shíjìnzhì	Another term for decimal (base-10)
28	Depth-first search	深度优先搜索	shēndù yōuxiān sōusuǒ	Explores as far as possible along each branch before backtracking
29	Distributed system	分布式系统	fēnbùshì xìtǒng	A system with multiple autonomous nodes cooperating
30	Domain name	域名	yùnmíng	A human-readable address for a website
31	Dynamic allocation	动态分配	dòngtài fēnpèi	Memory allocated at runtime
32	Edge computing	边缘计算	biānyuán jìsuàn	Processing data near the source rather than in a central cloud
33	Encryption	加密	jiāmì	Converting data into coded form to prevent unauthorised access
34	Entity	实体	shí tǐ	A distinct object or concept about which data is stored
35	Entity-relationship diagram	实体关系图	shí tǐ guān xì tú	A diagram showing entities and their relationships
36	Ethernet	以太网	yǐ tài wǎng	A widely used LAN technology
37	Exponent	指数	zhǐ shù	The power to which the base is raised
38	Feasibility study	可行性研究	kě xíng xìng yán jiū	Analysis of technical, economic, and operational viability
39	Fetch-decode-execute	取指-译码-执行	qǔ zhǐ - yì mǎ - zhí xíng	The CPU cycle of fetching, decoding, and executing instructions
40	Firewall	防火墙	fáng huǒ qiáng	A security system monitoring network traffic
41	Floating-point	浮点数	fú diǎn shù	A representation for real numbers with fractional parts
42	Flowchart	流程图	liú chéng tú	A diagram representing the sequence of operations

#	English	中文	Pinyin	Definition
43	Function	函数	hánshù	A sub-program that returns a value
44	Graph	图	tú	A set of vertices connected by edges
45	GUI	图形用户界面	túxíng yònghù jièmiàn	Graphical User Interface
46	Hashing	哈希	hāxī	Converting data to a fixed-size value using a hash function
47	Heap	堆	duī	A tree-based data structure satisfying the heap property
48	Hexadecimal	十六进制	shíliù jìnzhì	Base-16 number system (0 - 9, A - F)
49	HTML	超文本标记语言	chāo wénběn biāojiǔ yǔyán	HyperText Markup Language
50	HTTP	超文本传输协议	chāo wénběn chuánsū xiéyì	HyperText Transfer Protocol
51	HTTPS	安全超文本传输协议	ānquán chāo wénběn chuánsū xiéyì	HTTP Secure (encrypted)
52	Inorder traversal	中序遍历	zhōngxù biànlì	Left, root, right tree traversal
53	Input	输入	shūrù	Data entered into a system
54	Insertion sort	插入排序	chārù páixù	Builds sorted array one item at a time
55	Integer	整数	zhěngshù	A whole number (positive, negative, or zero)
56	Internet	互联网	hùliánwǎng	A global network of interconnected networks
57	Interrupt	中断	zhōngduàn	A signal suspending current processing
58	IP address	IP地址	IP dìzhǐ	A unique numerical identifier for a device
59	Iteration	迭代	diédài	Repeating a set of instructions
60	LAN	局域网	júyù wǎng	Local Area Network
61	Latency	延迟	yáncí	Time delay in data transmission
62	Linked list	链表	liànbiǎo	A dynamic data structure with nodes linked by pointers

#	English	中文	Pinyin	Definition
63	Logic gate	逻辑门	luójí mén	An electronic circuit performing a Boolean function
64	Lossless compression	无损压缩	wúsǔn yāsuō	Reduces file size without losing data
65	Lossy compression	有损压缩	yǒusǔn yāsuō	Reduces file size by permanently removing data
66	Machine code	机器码	jīqì mǎ	Low-level instructions executed directly by CPU
67	MAC address	MAC地址	MAC dìzhǐ	A unique hardware identifier for network interfaces
68	Mantissa	尾数	wěishù	The significant digits in a floating-point number
69	Memory	内存	nèicún	Hardware storing data and instructions
70	Merge sort	归并排序	guībìng páixù	A divide-and-conquer sorting algorithm
71	Microcontroller	微控制器	wēi kòngzhì qì	A compact integrated circuit for embedded control
72	Modem	调制解调器	tiáozhì jiětiáo qì	Converts digital data to analog signals
73	Network	网络	wǎngluò	A collection of connected computers
74	Node	节点	jiédiǎn	An element in a linked list or tree
75	Normalisation	规范化	guīfànhuà	Organising data to reduce redundancy
76	Operating system	操作系统	cāozuò xìtǒng	Software managing computer hardware and software
77	Overflow	溢出	yìchū	When a calculation exceeds storage capacity
78	Packet	分组	fēnzǔ	A unit of data transmitted over a network
79	Packet switching	分组交换	fēnzǔ jiāohuàn	Breaking data into packets sent independently
80	Paging	分页	fēnyè	Dividing memory into fixed-size pages

#	English	中文	Pinyin	Definition
81	Peer-to-peer	对等网络	duìděng wǎngluò	Network where nodes share resources directly
82	Pointer	指针	zhǐzhēn	A reference to a memory location
83	Postorder traversal	后序遍历	hòuxù biànlì	Left, right, root tree traversal
84	Preorder traversal	前序遍历	qiánxù biànlì	Root, left, right tree traversal
85	Primary key	主键	zhǔjiàn	A unique identifier for a database record
86	Procedure	过程	guòchéng	A sub-program that does not return a value
87	Process	进程	jìnchéng	An instance of a program in execution
88	Protocol	协议	xiéyì	A set of rules governing data communication
89	Pseudocode	伪代码	wěi dàimǎ	A language-independent description of an algorithm
90	Queue	队列	duìliè	A FIFO data structure
91	RAM	随机存取存储器	suíjī cúnqǔ cúnchǔ qì	Volatile primary memory
92	Recursion	递归	dìguī	A function calling itself
93	Record	记录	jìlù	A composite data type with multiple fields
94	Register	寄存器	jìcúnqì	High-speed storage within the CPU
95	Relationship	关系	guānxì	An association between entities
96	Router	路由器	lùróu qì	A device forwarding packets between networks
97	ROM	只读存储器	zhǐdú cúnchǔ qì	Non-volatile permanent memory
98	Round robin	轮询	lún xún	A scheduling algorithm giving each process equal time
99	Routing	路由	lùyóu	Selecting paths for traffic in a network
100	Search algorithm	搜索算法	sōusuǒ suànfǎ	A procedure for finding an item in a collection

#	English	中文	Pinyin	Definition
101	Secondary storage	辅助存储器	fǔzhù cúnchǔ qì	Non-volatile persistent storage
102	Segmentation	分段	fēnduàn	Dividing memory into variable-sized segments
103	Selection	选择	xuǎnzé	A control structure choosing between paths
104	Selection sort	选择排序	xuǎnzé páixù	Repeatedly finds minimum element
105	Semaphore	信号量	xìnhào liàng	A variable controlling access to shared resources
106	Sensor	传感器	chuángǎnqì	A device detecting physical properties
107	Server	服务器	fúwùqì	A computer providing services to clients
108	Software	软件	ruǎnjiàn	Programs and operating information
109	Sorting algorithm	排序算法	páixù suànfǎ	A procedure for arranging data in order
110	SQL	结构化查询语言	jiégòuhuà cháxún yǔyán	Structured Query Language
111	Stack	栈	zhàn	A LIFO data structure
112	Stakeholder	利益相关者	lìyì xiāngguān zhě	Anyone with interest in a system
113	State machine	状态机	zhuàngtài jī	A model of computation with states and transitions
114	String	字符串	zìfúchuàn	A sequence of characters
115	Subnet mask	子网掩码	zǐ wǎng yǎnmǎ	Divides IP address into network and host
116	Success criteria	成功标准	chénggōng biāozhǔn	Measurable conditions for project success
117	Symmetric encryption	对称加密	duìchèn jiāmì	Same key for encryption and decryption
118	Syntax	语法	yǔfǎ	Rules governing program structure
119	System	系统	xìtǒng	A set of interconnected components working together
120	TCP/IP	传输控制协议/网际协议	chuánshū kòngzhì xiéyì/wǎngjì xiéyì	Fundamental internet protocol suite

#	English	中文	Pinyin	Definition
121	Thread	线程	xiànréng	A lightweight sub-process
122	Throughput	吞吐量	tūn tǔ liàng	Amount of data processed in a given time
123	Token	令牌	lìngpái	A temporary key for authentication
124	Topology	拓扑结构	tuōpǔ jiégòu	The arrangement of a network
125	Trace table	追踪表	zhuīzōng biǎo	A table showing variable values at each step
126	Transmission Control Protocol	传输控制协议	chuánshū kòngzhì xiéyì	Reliable, ordered delivery protocol
127	Traversal	遍历	biànlì	Visiting each node in a tree/graph
128	Tree	树	shù	A hierarchical data structure with nodes
129	Truncation error	截断错误	jiéduàn cuòwù	Error from approximating a mathematical process
130	Two's complement	补码	bǔmǎ	Method for representing signed integers
131	Underflow	下溢	xià yì	When a number is too small to be represented
132	Unicode	统一码	tǒngyī mǎ	Standard for consistent text encoding
133	URL	统一资源定位符	tǒngyī zīyuán dìngwèi fú	Uniform Resource Locator
134	User interface	用户界面	yònghù jièmiàn	The point of interaction between user and system
135	Variable	变量	biànrliàng	A named storage location whose value can change
136	Virtual memory	虚拟内存	xūnǐ nèicún	Using disk space to extend apparent RAM
137	Virtual private network	虚拟专用网络	xūnǐ zhuānyòng wǎngluò	Secure private network over public network
138	Virus	病毒	bìngdú	Malicious software that replicates
139	WAN	广域网	guǎngyù wǎng	Wide Area Network
140	White-box testing	白盒测试	bái hé cèshì	Testing with knowledge of internal code

#	English	中文	Pinyin	Definition
141	Wi-Fi	无线保真	wúxiàn bǎo zhēn	Wireless networking technology
142	Word length	字长	zìcháng	Number of bits processed in one operation
143	XML	可扩展标记语言	kě kuòzhǎn biāojiǔ yǔyán	eXtensible Markup Language
144	XOR	异或	yì huò	Exclusive OR logic gate

Topic-Specific Term Lists

For focused revision, use these topic groupings:

Topic 1: System Fundamentals

Stakeholder, feasibility study, changeover, entity, relationship, cardinality, flowchart, IPO table, milestone, system, user-centered design

Topic 2: Computer Organization

CPU, ALU, control unit, register, bus, RAM, ROM, cache, virtual memory, two's complement, logic gate, mantissa, exponent, interrupt, word length

Topic 3: Networks

LAN, WAN, protocol, packet switching, router, IP address, DNS, encryption, firewall, VPN, TCP/IP, Ethernet, topology, bandwidth, latency

Topic 4: Computational Thinking

Pseudocode, variable, array, iteration, recursion, algorithm, efficiency, trace table, black-box testing, white-box testing, procedure, function

Topic 5: Abstract Data Structures (HL)

Linked list, node, pointer, binary tree, binary search tree, traversal, graph, adjacency matrix, depth-first search, breadth-first search, dynamic allocation, heap

Topic 6: Resource Management (HL)

Process, thread, context switch, paging, segmentation, deadlock, semaphore, scheduling, round robin, memory management, virtual memory

Topic 7: Control (HL)

Centralized control, distributed system, consensus, autonomous agent, emergent behaviour, CAP theorem, failover, swarm intelligence, edge computing, algorithmic bias

General / Cross-Topic

Algorithm, binary, byte, compression, encryption, GUI, HTML, HTTP, HTTPS, internet, normalisation, SQL, URL, XML, syntax, topology, user interface

Student Glossary Checklist:

- ☐ I have created flashcards for all 180 terms
- ☐ I can define each term without looking
- ☐ I can write an example sentence for each term
- ☐ I know the Chinese translation for key terms
- ☐ I review my flashcards at least twice per week
- ☐ Victoria Park Academy Shenzhen | IB DP Computer Science | Bilingual Glossary